Robustness and explainability for complex data objects

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1 Introduction and overview

The field of robust statistics is closely related to the development of methods for outlier detection [Maronna et al., 2019]. Generally, outliers are considered observations that deviate to some extent from the main bulk of the data. In fact, there are approaches from two communities: the statistics community, usually assuming that outliers are generated from an unspecified distribution that differs from the assumed distribution of the main data group [Rousseeuw et al., 2005], and the machine learning community, mostly trying to identify isolated observations, without taking distributional assumptions [Zimek and Filzmoser, 2018]. Our focus is on a statistical approach that utilizes specific model assumptions. In the multivariate setting, it is common to assume a multivariate normal distribution as the data-generating process, with mean $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)'$ and a $p \times p$ covariance matrix $\boldsymbol{\Sigma}$. A widely used outlyingness measure of an observation $\boldsymbol{x} = (x_1, \ldots, x_p)'$ is the (squared) Mahalanobis distance (MD), defined as

$$\mathrm{MD}^2_{\boldsymbol{\mu},\boldsymbol{\Sigma}}(\boldsymbol{x}) = (\boldsymbol{x} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}) \tag{1}$$

[Mahalanobis, 1936]. The parameters $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ have to be robustly estimated based on n available observations. Several proposals exist in the literature, and one of the most popular estimators is the Minimum Covariance Determinant (MCD) estimator [Rousseeuw and Driessen, 1999]. For a given fixed number h, which is roughly half of the sample size n, its objective is to select h (out of n) observations for which the calculated sample covariance matrix has the lowest determinant. The robust covariance estimator is given by the sample covariance matrix of the h observations leading to the minimum, multiplied by a consistency factor, and the estimate of the mean is given by the arithmetic mean of those h observations [Rousseeuw, 1985]. An appropriate cutoff value [Rousseeuw and Driessen, 1999, Maronna et al., 2019] enables us to distinguish outliers from regular observations. The more flexible proposal of Schreurs et al. [2021] introduces a kernel-based version of the MCD estimator for outlier detection for non-elliptical data.

Outlier detection became more challenging in the high-dimensional setting, especially for $n < p$, since Mahalanobis distances cannot be computed based on a singular covariance estimate. As a way out, projection-based procedures [Filzmoser et al., 2008] and versions based on a regularized covariance matrix [Boudt et al., 2020] or a precision matrix [Öllerer and Croux, 2015, Tarr et al., 2016] were introduced.

With increasing technological advances, measurements became more complex, leading to data objects that are not just multivariate vectors, but matrices, tensors, higher-order (multi-way) tensors, functions, multivariate functions, or higher-order functional objects (surfaces, etc.). While vectorization in the matrix/tensor setting or discretization in the functional setting makes it possible to still use the standard tools developed for multivariate data, important information about the structure of these objects gets lost. Recently, we introduced robust methods, also in the context of outlier detection, for some of these complex data objects:

- Mayrhofer et al. [2025a] introduced robust covariance estimation and explainable outlier detection for matrix-valued data;
- first steps towards outlier detection for tensor-valued data were made in the master's thesis Obukhov [2025];

- in Oguamalam et al. [2024b], we introduced an outlier detection method based on a regularized Mahalanobis distance for univariate functional data;
- in the PhD thesis Mayrhofer [2024], we introduced a truncated Mahalanobis distance for outlier detection of uni- and multivariate functional data.

We will be more specific in subsequent sections regarding the methods and related literature, but these works are just the beginning of a more fundamental research of robust methodology for more complex data objects.

Another equally important and related topic is explainability. In the context of outlier detection, we would like to know what makes an observation an outlier. To answer this question, we have developed Shapley values for outlier explanation [Mayrhofer and Filzmoser, 2023]. Shapley values are known from game theory [Shapley, 1953] and are widely used in Explainable AI [Lundberg and Lee, 2017]. They are particularly interesting for interpretable outlier detection, since they additively decompose a squared (robust) MD of a multivariate observation into contributions originating from single variables [Mayrhofer and Filzmoser, 2023]. Thus, a high contribution related to a variable means that this variable has a strong effect on the outlyingness of the observation. There are several other attractive properties of Shapley values [see Young, 1985], and one practical advantage of using Shapley values in this context is that they can be computed with linear complexity in p [Mayrhofer and Filzmoser, 2023]. Indeed, this contrasts with Shapley values in the context of Explainable AI, where a combinatorial problem needs to be solved, leading to a high computational effort [Lundberg and Lee, 2017].

In the paper Mayrhofer et al. [2025a], we managed to extend Shapley values for outlier explanation to the matrix-valued setting. In Mayrhofer [2024], we transferred the concept of outlier explanation to the functional setting. We consider these works as a crucial step towards explainability in robust statistics, as it marks the beginning of a well-founded development in this direction.

We believe that our work listed above and experience with robust and explainable outlier detection for more complex data objects is an ideal basis to continue this research with new and challenging topics:

The **project's main objective** is to develop robust and explainable methods for classification and clustering for complex data objects, i.e. for tensor-valued data and higher-order tensors, and for uni- and multivariate functional data. We will focus on the concept of Mahalanobis distance, deriving novel robust and regularized versions in these settings, which also work for small sample sizes and high dimensions. The developed methodology will fill a gap, as complex data objects are commonly vectorized or discretized. Moreover, robust explainability in classification and clustering has not been considered before.

2 Project goals, innovation, and originality

Robust regularized Mahalanobis distances will be the basis for robust classification and clustering, as well as for explainability. In this context, this is a new concept for complex data objects. The way in which Mahalanobis distances can be derived for the data types we are considering varies greatly and divides our research tasks. We can formulate the following hypotheses and objectives for our project:

(Hyp. I) **Robust regularized Mahalanobis distances can be used for explainable classification and clustering of tensor-valued (and higher-order) observations, also in higher dimensions.**

- (Obj. I.1) Derive theoretical properties of the estimators.
- (Obj. I.2) Develop tools for outlier explanation.
- (Obj. I.3) Extend the concepts to multi-way tensors.
- (Obj. I.4) Develop methodology for regularized covariance estimation to cope with high-dimensional, low sample size settings.
- (Obj. I.5) Develop distance-based classification and clustering methods.
- (Obj. I.6) Develop tools to obtain explainability for classification and clustering, including visualizations based on dimension reduction.

Section 4 will provide more details on robustness for tensor-valued data, and the research objectives will be linked to a work package and to specific tasks.

(Hyp. II) **Robust regularized Mahalanobis distances can be defined for multivariate functional data and used for explainable classification and clustering.**

- (Obj. II.1) Develop a regularized Mahalanobis distance.
- (Obj. II.2) Develop robust covariance estimators for separable and non-separable models.
- (Obj. II.3) Develop distance-based classification and clustering methods.
- (Obj. II.4) Develop tools to obtain explainability for classification and clustering.
- (Obj. II.5) Derive theoretical properties of the estimators.

More details on the use of Mahalanobis distances for multivariate functional data are provided in **Section** 5. There, we also link the research objectives to a work package and to specific tasks.

The presented research topics are innovative and original:

- *Robust explainable classification:* For vector-valued data, LDA (linear discriminant analysis) uses a pooled covariance estimate to avoid over-parametrization. What would this mean in the matrix/tensor valued case, and how to do (robust) pooling in the functional case? What does explainability refer to for complex data objects? Can we identify for each observation which cell of a matrix/tensor or which subdomain of a (multivariate) function leads to the classification into a particular group? We believe that answers to these questions can open up entirely new avenues in classification.
- *Robust explainable clustering:* Restrictions on the covariances need to be imposed in order to avoid degenerated solutions [e.g. García-Escudero et al., 2022]. It is very challenging to define appropriate constraints for more complex data objects. Explainability (which sub-information leads to the assignment of an observation to a cluster) is also unexplored, even for the vector-valued case. With these innovations, we expect outstanding contributions to the field of cluster analysis.

- *Small sample size n in combination with large dimensionality* is a challenging problem already in the vector-valued case. A way out is regularization, but it remains unclear what the best way could be to regularize Mahalanobis distances and/or covariance estimates. We made a proposal for univariate functional data [Oguamalam et al., 2024b], and simply used truncation for multivariate functions [Mayrhofer, 2024], but developing a proper regularization strategy would indeed break new ground.

The availability of robust classification and clustering methods for complex data types will have a great impact on many applied fields. In particular, tools for explainability will be helpful for practitioners to gain deeper insight into the results. The developed methodology has great potential for further innovative research, for example, on robust explainable regression.

3 Matrix-valued observations: state-of-the-art

In this section we briefly explain the concept of Shapley values [Shapley, 1953] for outlier explanation, as introduced in Mayrhofer and Filzmoser [2023], since this is essential for the whole proposal. We then describe outlier detection and explanation for matrix-valued observations [Mayrhofer et al., 2025a], which will be helpful to understand the approaches for the other data types treated in this proposal.

3.1 Shapley values for outlier explanation

Consider an observation vector $\boldsymbol{x} = (x_1, \ldots, x_p)'$ from a population mean $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_p)'$ and covariance matrix $\boldsymbol{\Sigma}$. We would like to investigate the contribution of the j-th coordinate x_j to the outlyingness of $\boldsymbol{x}$. Shapley values, as they are known from game theory, assume a set of players $P = \{1, \ldots, p\}$, which are here the individual variable indices [Shapley, 1953]. Denote by S a coalition of players, thus a subset of P. In cooperative game theory, the goal is to determine a fair payoff distribution among the players. Mayrhofer and Filzmoser [2023] define the payoff function in terms of the squared Mahalanobis distance, see Equation (1), as $\mathrm{MD}^2_{\boldsymbol{\mu},\boldsymbol{\Sigma}}(\boldsymbol{x}^S) = \mathrm{MD}^2(\boldsymbol{x}^S)$, with $\boldsymbol{x}^S = (x_1^S, \ldots, x_p^S)'$,

$$x_j^S := \begin{cases} x_j & \text{if } j \in S \\ \mu_j & \text{if } j \notin S \end{cases}, \tag{2}$$

which fulfills $\mathrm{MD}^2(\boldsymbol{x}^S) = 0$, if $S = \emptyset$ is the empty set. Thus, if the j-th variable is not part of the coalition, its coordinate value is set to the overall coordinate mean. The marginal contribution of the k-th variable is given as $\Delta_k \mathrm{MD}^2(\boldsymbol{x}^S) := \mathrm{MD}^2(\boldsymbol{x}^{S\cup\{k\}}) - \mathrm{MD}^2(\boldsymbol{x}^S)$, thus evaluating the difference in the payoff function by including or excluding variable k from the coalition. The Shapley value aggregates over all possible 2^{p-1} subsets $S \subseteq P \setminus \{k\}$ and is given by

$$\phi_k(\boldsymbol{x}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{S \subseteq P\setminus\{k\}} \frac{|S|!(p - |S| - 1)!}{p!} \Delta_k \mathrm{MD}^2(\boldsymbol{x}^S) = (x_k - \mu_k) \sum_{j=1}^{p} (x_j - \mu_j)\omega_{jk}, \tag{3}$$

with $\boldsymbol{\Sigma}^{-1} =: \boldsymbol{\Omega} = (\omega_{jk})_{j,k=1,\ldots,p}$ and $\boldsymbol{x}^S$ as in Equation (2). The simplification to Equation (3) is proven in Mayrhofer and Filzmoser [2023], and suggests linear complexity in p.

Shapley values have several desirable properties, such as symmetry, monotonicity, and efficiency; see Young [1985]. The property of efficiency is particularly relevant to the outlier explanation, as it

states that the sum of the Shapley values for all variables equals the squared MD, i.e.

$$\sum_{j=1}^{p} \phi_j(\boldsymbol{x}) = \mathrm{MD}^2(\boldsymbol{x}). \tag{4}$$

Due to their construction, Shapley values can also be negative; see Mayrhofer and Filzmoser [2023] for details on their interpretation. In addition to outlier explanation, this concept relates to cellwise outlyingness [Raymaekers and Rousseeuw, 2024], where Shapley values provide variable contributions for outlying observations relative to their "expected" position within the multivariate structure [Mayrhofer and Filzmoser, 2023].

3.2 Outlier detection for matrix-valued observations

Consider a random variable $\boldsymbol{X}$, with p rows and q columns, with a matrix elliptical distribution. The matrix elliptical family is a semi-parametric class of distributions parametrized by the mean $\mathbf{M} \in \mathbb{R}^{p\times q}$, row covariance $\boldsymbol{\Sigma}^{\mathrm{row}} \in \mathrm{PDS}(p)$, column covariance $\boldsymbol{\Sigma}^{\mathrm{col}} \in \mathrm{PDS}(q)$, and the so-called density generator function $g : [0, \infty) \to \mathbb{R}$. Here, $\mathrm{PDS}(a)$, with $a \in \mathbb{N}$, denotes the class of all positive definite symmetric $a \times a$ matrices. More specifically, a random matrix $\boldsymbol{X}$ with an absolutely continuous distribution has an elliptical distribution, denoted $\mathcal{ME}(\mathbf{M}, \boldsymbol{\Sigma}^{\mathrm{row}}, \boldsymbol{\Sigma}^{\mathrm{col}}, g)$, if its density can be written as

$$f(\boldsymbol{X}) = \det(\boldsymbol{\Sigma}^{\mathrm{row}})^{-q/2} \det(\boldsymbol{\Sigma}^{\mathrm{col}})^{-p/2} g(\mathrm{tr}(\boldsymbol{\Omega}^{\mathrm{col}}(\boldsymbol{X} - \mathbf{M})'\boldsymbol{\Omega}^{\mathrm{row}}(\boldsymbol{X} - \mathbf{M}))), \tag{5}$$

with $\boldsymbol{\Omega}^{\mathrm{row}} = (\boldsymbol{\Sigma}^{\mathrm{row}})^{-1}$ and $\boldsymbol{\Omega}^{\mathrm{col}} = (\boldsymbol{\Sigma}^{\mathrm{col}})^{-1}$ denoting the precision matrices among the rows and columns, respectively. Matrix elliptical distributions can also be related to their multivariate counterparts. Formally, a random matrix $\boldsymbol{X}$ follows a matrix elliptical distribution $\mathcal{ME}(\mathbf{M}, \boldsymbol{\Sigma}^{\mathrm{row}}, \boldsymbol{\Sigma}^{\mathrm{col}}, g)$ if and only if its vectorized version $\mathrm{vec}(\boldsymbol{X})$ follows a multivariate elliptical distribution $\mathcal{E}(\mathrm{vec}(\mathbf{M}), \boldsymbol{\Sigma}^{\mathrm{col}} \otimes \boldsymbol{\Sigma}^{\mathrm{row}}, g)$ [Gupta and Varga, 2012]. Here, $\mathrm{vec}(\cdot)$ is the vectorization operator, stacking the columns of a matrix on top of each other, and $\otimes$ is the Kronecker product. Densities from the semi-parametric family of matrix elliptical distributions depended on $\boldsymbol{X}$ only via the squared matrix Mahlanobis distance (MMD), defined as

$$\mathrm{MMD}^2(\boldsymbol{X}) := \mathrm{MMD}^2(\boldsymbol{X}; \mathbf{M}, \boldsymbol{\Sigma}^{\mathrm{row}}, \boldsymbol{\Sigma}^{\mathrm{col}}) = \mathrm{tr}(\boldsymbol{\Omega}^{\mathrm{col}}(\boldsymbol{X} - \mathbf{M})'\boldsymbol{\Omega}^{\mathrm{row}}(\boldsymbol{X} - \mathbf{M})), \tag{6}$$

making MMD a sufficient statistic within this family.

Probably the most studied matrix elliptical distribution is the matrix normal distribution [Dawid, 1981], denoted $\mathcal{MN}(\mathbf{M}, \boldsymbol{\Sigma}^{\mathrm{row}}, \boldsymbol{\Sigma}^{\mathrm{col}})$, with density

$$f(\boldsymbol{X} | \mathbf{M}, \boldsymbol{\Sigma}^{\mathrm{row}}, \boldsymbol{\Sigma}^{\mathrm{col}}) = \frac{\exp(-\frac{1}{2}\mathrm{tr}(\boldsymbol{\Omega}^{\mathrm{col}}(\boldsymbol{X} - \mathbf{M})'\boldsymbol{\Omega}^{\mathrm{row}}(\boldsymbol{X} - \mathbf{M})))}{(2\pi)^{pq/2} \det(\boldsymbol{\Sigma}^{\mathrm{col}})^{p/2} \det(\boldsymbol{\Sigma}^{\mathrm{row}})^{q/2}}. \tag{7}$$

The classical estimators of the parameters $\mathbf{M}$, $\boldsymbol{\Sigma}^{\mathrm{col}}$, and $\boldsymbol{\Sigma}^{\mathrm{row}}$ are found by maximum likelihood estimation [Dutilleul, 1999]. In Mayrhofer et al. [2025a] it was shown that robust estimators can be obtained just in the spirit of the MCD estimator, through a weighted maximum likelihood problem, by defining weights $w_i \in \{0, 1\}$ for the observations $i \in \{1, \ldots, n\}$, such that $\sum_{i=1}^{n} w_i = h$, with $n/2 \leq h \leq n$, where n is the number of matrix-valued observations. The resulting MMCD (matrix MCD) estimators have several desirable properties, such as matrix affine equivariance, high breakdown

point, and consistency under elliptical distributions [see Mayrhofer et al., 2025a]. Also, an algorithm to efficiently compute these estimators has been proposed in Mayrhofer et al. [2025a].

Once the robust estimators are available, thus $\hat{\mathbf{M}}$, $\hat{\mathbf{\Sigma}}^{\text{row}}$, and $\hat{\mathbf{\Sigma}}^{\text{col}}$, an outlier detection rule can be implemented as

$$\text{MMD}^2(\boldsymbol{X}_i; \hat{\mathbf{M}}, \hat{\mathbf{\Sigma}}^{\text{row}}, \hat{\mathbf{\Sigma}}^{\text{col}}) > \chi^2_{pq;0.975} \ ,$$

for a matrix-valued observation $\boldsymbol{X}_i$, $i \in \{1, \ldots, n\}$, and the MMCD estimators $\hat{\mathbf{M}}$, $\hat{\mathbf{\Sigma}}^{\text{row}}$, and $\hat{\mathbf{\Sigma}}^{\text{col}}$.

In practice, also in machine learning, matrix-variate observations are simply vectorized to make traditional algorithms applicable. Thus, matrix-variate observations with p rows and q columns are converted to vectors of length $p \cdot q$, which can be very high-dimensional, leading to severe limitations of many methods and usually to the need for many observations. In the case of $p \approx q$, even if they are high, the MMCD procedure requires just a couple of observations to compute robust estimators [Mayrhofer et al., 2025a]. This reduction in the required sample size is due to the restricted covariance structure which reduces the number of parameters to be estimated.

3.3 Outlier explanation for matrix-valued observations

Given a matrix sample $\boldsymbol{X} \in \mathbb{R}^{p \times q}$ with a mean matrix $\mathbf{M}$ and row and column covariances $\mathbf{\Sigma}^{\text{row}}$ and $\mathbf{\Sigma}^{\text{col}}$, respectively, Shapley values can be used to quantify the contribution of each cell in $\boldsymbol{X}$ to its overall deviation from the mean. By vectorizing the matrix, the problem is transformed into a multivariate setting, where the covariance matrix is expressed using the Kronecker product: $\mathbf{\Sigma} = \mathbf{\Sigma}^{\text{col}} \otimes \mathbf{\Sigma}^{\text{row}}$.

The Shapley values for each cell can be computed using the following expression [Mayrhofer et al., 2025a],

$$\phi_{jk}(\boldsymbol{X}) = (x_{jk} - m_{jk}) \sum_{i=1}^{p} \sum_{l=1}^{q} (x_{il} - m_{il}) \omega_{ij}^{\text{row}} \omega_{kl}^{\text{col}},$$

where ω_{ij}^{row} and ω_{kl}^{col} are components of the corresponding weight matrices. These values, organized into a matrix $\mathbf{\Phi}(\boldsymbol{X})$, provide a detailed view of how much each cell in $\boldsymbol{X}$ contributes to its overall distance from the mean in the sense $\mathbf{\Phi}(\boldsymbol{X}) = (\boldsymbol{X} - \mathbf{M}) \circ \mathbf{\Omega}^{\text{row}} (\boldsymbol{X} - \mathbf{M}) \mathbf{\Omega}^{\text{col}}$, where $\circ$ denotes element-wise multiplication. This factorization enables efficient computation of the cell-wise Shapley values of $\boldsymbol{X}$.

The Shapley values for matrix-variate data exhibit several important properties under transformations [Mayrhofer et al., 2025a]. They are invariant to shifts, meaning that adding a constant matrix $\boldsymbol{C}$ does not change the Shapley values: $\mathbf{\Phi}(\boldsymbol{X} + \boldsymbol{C}) = \mathbf{\Phi}(\boldsymbol{X})$. They are also scale invariant when scaling matrices (diagonal matrices with non-zero entries) are applied. Furthermore, they are permutation equivariant, meaning that rearranging rows or columns of $\boldsymbol{X}$ results in a corresponding rearrangement of the Shapley values. For example, mirroring or rotating a matrix, such as a gray-scale image, would result in a similar transformation of the Shapley values. However, Shapley values are not affine equivariant in the general sense, as arbitrary linear transformations do not preserve their structure.

In addition to cellwise contributions, assessing Shapley values at the row or column level can be helpful. This is particularly relevant in settings where comparisons across observations or features are of interest. The rowwise Shapley values, for instance, can be computed by summing the contributions of all cells within each row,

$$\phi_{j\cdot}(\boldsymbol{X}) = \sum_{k=1}^{q} \phi_{jk}(\boldsymbol{X}),$$

leading to a vector of rowwise Shapley values: $\boldsymbol{\phi}_{\text{row}}(\boldsymbol{X}) = \text{diag}(\boldsymbol{\Omega}^{\text{row}}(\boldsymbol{X} - \mathbf{M})\boldsymbol{\Omega}^{\text{col}}(\boldsymbol{X} - \mathbf{M})')$. Similarly, columnwise Shapley values can be derived using analogous computations. These aggregated values retain the same invariance properties as the cellwise Shapley values, making them a valuable tool for analyzing contributions at different levels of granularity. E.g., for repeated multivariate measurement datasets, row-wise Shapely values can be used to detect overall anomalous measurements for an individual, while the column-wise Shapley value indicates which measured attribute is deviating. Figure 1 illustrates the use of rowwise Shapley values applied to monthly weather data from Austria's highest weather station "Hoher Sonnblick" (data source: GeoSphere Austria - `https://data.hub.geosphere.at`). The observed parameters are monthly averages of temperature (T), sunshine hours (SH), precipitation (P), air pressure (AP), and proportion of solid precipitation (SP), from 1891 to 2022. Outliers have occurred more frequently in recent years, and the figure indicates the reasoning; see Mayrhofer et al. [2025a] for details.

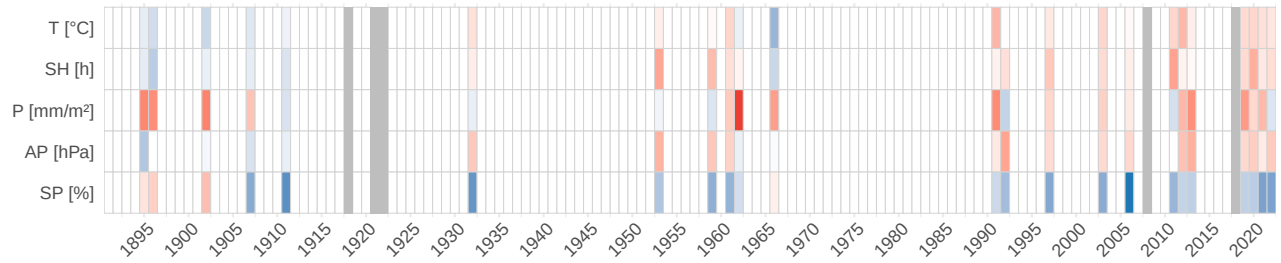


Figure 1: Yearly outlyingness contributions for the glacier weather data. Regular years are white, and years with missing data are gray. Outliers are colored as follows: blue for "below average", red for "above average", and color intensity proportional to the rowwise Shapley value.

In Figure 2, we demonstrate an example of cell-wise Shapley values for gray-scale video data analysis using a 633-frame video that depicts a beach. From frame 480 onward, a man walks across the scene, partially obscured by a tree; see Mayrhofer et al. [2025a] for details. The figure shows 4 of the frames identified as outliers, with red-colored pixels indicating increased Shapley values. Thus, this information immediately explains the reason for the outlyingness of the frames.

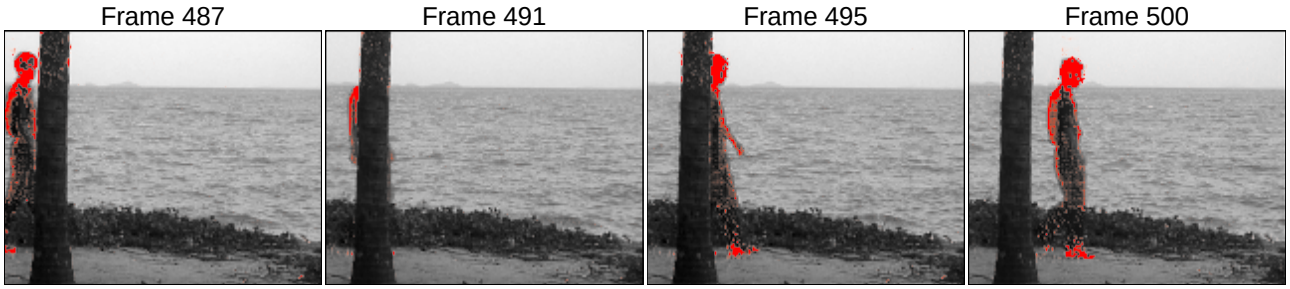


Figure 2: Outlyingness contribution based on cellwise Shapley values are shown in red, where darker colors indicate higher outlyingness contributions. The grayscale video frames are displayed in the background.

4 Tensor-valued observations: new developments

While matrix-valued observations are reported with a certain number of rows and columns (e.g., defined by the resolution of an image), tensor-valued observations have additional modes. Here, we primarily

refer to one additional mode, resulting in a three-way tensor, also known as a third-order tensor, which consists of observations with p_1 rows, p_2 columns, and p_3 slices. Thus, the tensor-valued observation consists of p_3 matrices. An example is RGB video data, where the video information is recorded in terms of $p_3 = 3$ matrices, each with $p_1 \times p_2$ pixels, for n frames. In the example shown in Figure 2, we transformed the RGB video data into gray-scale images to obtain matrix-variate observations. Preferably, we also want to model the dependence structure between the color channels and treat each frame as a three-way tensor; see Figure 3.

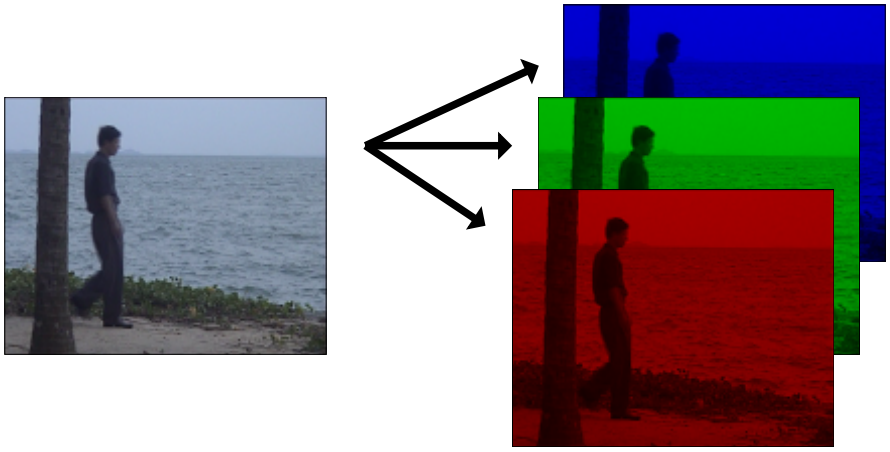


Figure 3: Treating a frame of an RGB video as a three-way tensor with red, green, and blue color information stored in the 3 slices of a tensor.

Similarly to the matrix-valued case, the goal is to robustly estimate the location and covariance for tensor-valued data by accounting for the inherent data structure based on a weighted likelihood formulation. In particular, we consider the problem of robust covariance estimation within the semi-parametric family of elliptically contoured tensors [Llosa-Vite and Maitra, 2024]. Like in the matrix case, the tensor normal distribution is among its most studied members. For the sake of conciseness and simplicity of the notation, in this proposal, we focus on the tensor normal distribution for a third-order tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{p_1 \times p_2 \times p_3}$, denoted as $\boldsymbol{\mathcal{X}} \sim \mathcal{TN}(\boldsymbol{\mathcal{M}}; \boldsymbol{\Sigma}^1, \boldsymbol{\Sigma}^2, \boldsymbol{\Sigma}^3)$, where $\boldsymbol{\mathcal{M}}$ is a $p_1 \times p_2 \times p_3$ mean tensor, and $\boldsymbol{\Sigma}^1 \in \mathbb{R}^{p_1 \times p_1}$, $\boldsymbol{\Sigma}^2 \in \mathbb{R}^{p_2 \times p_2}$, $\boldsymbol{\Sigma}^3 \in \mathbb{R}^{p_3 \times p_3}$ are positive definite covariance matrices associated with each dimension. Note that in the literature, the tensor normal distribution is also called multilinear [Ohlson et al., 2013], or multiway [Wang et al., 2022], or array variate [Akdemir and Gupta, 2011] normal. $\boldsymbol{\mathcal{X}}$ follows this distribution if and only if $\text{vec}(\boldsymbol{\mathcal{X}}) \in \mathbb{R}^{p_1 p_2 p_3}$ follows a multivariate normal distribution $\mathcal{N}(\text{vec}(\boldsymbol{\mathcal{M}}), \boldsymbol{\Sigma}^3 \otimes \boldsymbol{\Sigma}^2 \otimes \boldsymbol{\Sigma}^1)$ [Manceur and Dutilleul, 2013]. Here, the operator $\text{vec}(\cdot)$ stacks the entries $x_{j_1 j_2 j_3}, j_k = 1, \ldots, p_k, k \in \{1, 2, 3\}$ of $\boldsymbol{\mathcal{X}}$ such that the first dimension varies the fastest, followed by the second, and so on, i.e., $\text{vec}(\cdot) = (x_{111}, x_{211}, \ldots, x_{p_1 11}, x_{121}, \ldots, x_{p_1 p_2 p_3}) \in \mathbb{R}^{p_1 p_2 p_3}$.

In the case of the tensor normal model, maximum likelihood estimation (MLE) of $\boldsymbol{\mathcal{M}}, \boldsymbol{\Sigma}^1, \boldsymbol{\Sigma}^2, \boldsymbol{\Sigma}^3$ can be performed via an iterative three-stage algorithm [Manceur and Dutilleul, 2013]. This approach generalizes the matrix normal MLE, also known as the "flip-flop" algorithm. For n observed tensors $\boldsymbol{\mathcal{X}}_1, \ldots, \boldsymbol{\mathcal{X}}_n$, the MLE problem aims at finding the parameters that maximize the log-likelihood function

$$\begin{aligned}\ell(\boldsymbol{\mathcal{M}}, \boldsymbol{\Sigma}^1, \boldsymbol{\Sigma}^2, \boldsymbol{\Sigma}^3) = &- \frac{n p_1 p_2 p_3}{2} \log(2\pi) - \frac{n}{2} \log(\det(\boldsymbol{\Sigma}^3 \otimes \boldsymbol{\Sigma}^2 \otimes \boldsymbol{\Sigma}^1)) \\ &- \frac{1}{2} \sum_{i=1}^{n} \text{vec}(\boldsymbol{\mathcal{X}}_i - \boldsymbol{\mathcal{M}})' (\boldsymbol{\Sigma}^3 \otimes \boldsymbol{\Sigma}^2 \otimes \boldsymbol{\Sigma}^1)^{-1} \text{vec}(\boldsymbol{\mathcal{X}}_i - \boldsymbol{\mathcal{M}}),\end{aligned}$$

where the last line contains the tensor Mahalanobis distances. The parameters can be estimated in a similar manner to the matrix-valued case, by utilizing a "flip-flop" scheme [Manceur and Dutilleul, 2013]. A major difficulty, however, is the evaluation of Kronecker products, as they result in a significant increase in dimensions. In the thesis Obukhov [2025], we have worked out a scheme where the dimension increase is avoided. Moreover, an MCD-alike procedure has been implemented to compute robust estimators of the parameters, which also leads to robustified tensor Mahalanobis distances as a basis for outlier detection.

The following work package WP I addresses the research question formulated as hypothesis Hyp. I in Section 2. In particular, the objectives mentioned in Section 2, Obj. I.1–I.6, are addressed in the following tasks:

WP I **Robust regularized Mahalanobis distances for explainable classification and clustering of tensor-valued and higher-order observations**

Objectives: The purpose of this work package is to develop outlier-robust clustering and classification methodology for third-order tensor-valued data and for data of higher order. This includes the definition of estimators, the derivation of their properties, the implementation of efficient algorithms, and the development of tools for explainability.

Innovation: Robust regularized estimators are a new and promising concept for tensors, and their use within Mahalanobis distances can be extremely powerful for clustering and classification with small sample sizes. Explainability for classification and clustering is also new and can be particularly insightful for tensor data.

(Task I.1) **Definition and theoretical properties of the estimators.** Starting from the tensor normal model, robust estimators of the parameters are defined, and their robustness properties, such as breakdown point and consistency, are derived. Efficient algorithms for their computation are developed. We can profit from our work [Mayrhofer et al., 2025a] for the matrix-valued case, where we were able to derive theoretical properties of the estimators, such as breakdown point and consistency for elliptical distributions.

(Task I.2) **Tools for outlier explanation.** An outlier explanation methodology is developed, including visualization tools to reveal the outlyingness information in a simple and effective way, leading to a contribution on the observation level, but also on the level of the modes (rows, columns, slices). Theoretical properties of the Shapley values are derived, in particular with respect to transformations. Here, our experience in the matrix-variate case [Mayrhofer et al., 2025a] can serve as a basis, but the extension is not straightforward.

(Task I.3) **Extension to multi-way tensors.** The developed three-way tensor version for outlier detection and explanation is generalized to higher-order objects. Robust Mahalanobis distances are estimated for higher-order objects, and Shapley values are used for their additive decomposition. As computation time can increase significantly, especially with higher dimensions, the design of efficient algorithms is a major challenge. Also an effective way of visualizing the outcome is non-trivial.

(Task I.4) **Develop methodology for regularized covariance estimation to cope with high-dimensional low sample size settings.** Regularization had been introduced for the MCD estimator in high-dimensional, low-sample settings [Boudt et al., 2020]. This was even

more crucial in a matrix- or tensor-valued setting, where many more parameters needed to be estimated. Recent promising work in this direction existed for the matrix-variate case [Zhang et al., 2023]. The authors proposed regularized covariance estimation methods for high-dimensional matrix-valued data under the condition of separability and a bandable covariance structure. Extending these ideas to the tensor-valued setting is highly challenging but constitutes a major methodological contribution, as no such framework had existed before. In addition, alternative regularization strategies popular in Machine Learning, such as nuclear–norm–based approaches [Yu and Kong, 2025], are incorporated into the tensor framework. Together, these developments yield a novel class of regularized tensor-valued covariance estimators tailored to high-dimensional robust analysis.

(Task I.5) **Develop distance-based classification and clustering methods.** Distances played a central role in many multivariate techniques, including discriminant analysis, k-nearest neighbors, and a wide range of clustering methods, which rely on the data only via the chosen distance function to assign observations to one of K classes; see Hastie et al. [2009]. One particularly important choice of distance is the Mahalanobis distance, which extends the Euclidean distance by accounting for the correlation structure within groups and was therefore especially useful for complex, structured data. Our goal is to develop robust and scalable distance-based classification and clustering methods for high-dimensional tensor-valued data. This is achieved by integrating the newly developed regularized covariance estimators into Mahalanobis-type distances and by adapting existing distance-based algorithms to operate reliably in the tensor setting. These developments lead to a coherent framework for robust classification and clustering that fully exploits the structure of multiway data while remaining computationally feasible in high dimensions.

(Task I.6) **Explainability for clustering and classification.** Whether an observation was assigned to a group could depend on different modes of the matrix or tensor, as well as on specific single-cell entries. We pair Mahalanobis-type distances with adapted Shapley values to quantify feature contributions to the assignment. In particular, the explainability framework is expanded to show not only why an observation was assigned to a specific cluster, but also why it was not assigned to other clusters. Additionally, we develop methods to identify which tensor modes or features drive the separation between clusters, providing insight into the underlying structure of the data. Visualizations are created to support these explanations, including robust dimension-reduction methods. For that purpose, robust PCA is used by incorporating robust covariance estimators as plug-ins to robustify the PCA, following ideas similar to Hirari et al. [2025].

5 Multivariate functional data: state-of-the-art and new developments

In many real-world applications, data take the form of functions or curves recorded over time, space, or another continuous domain. Outliers in these functional observations can significantly influence the results of statistical analyses, highlighting the need for specialized techniques in Functional Data Analysis (FDA).

5.1 State-of-the-art

Mathematical framework: Let $(\Omega, \mathcal{A}, P)$ be a probability space and $\mathcal{T} \subset \mathbb{R}$ a compact interval, commonly thought of as time, then $\boldsymbol{X} = \{\boldsymbol{X}(t,\omega), t \in \mathcal{T}\} = \{(X_1(t,\omega), \ldots, X_p(t,\omega))', t \in \mathcal{T}\} : \mathcal{T} \times \Omega \to \mathbb{R}^p$ is a time-continuous vector-valued stochastic process. Hence, $\boldsymbol{X}$ is a collection of random variables defined on a common probability space index by the continuous set $\mathcal{T}$; for every fixed $t \in \mathcal{T}$ the process defines a random variable $\boldsymbol{X}(t)$, and for every fixed $\omega \in \Omega$ a sample path or trajectory $\boldsymbol{X}(.,\omega)$, i.e., a function of $t \in \mathcal{T}$. Each realization of $\boldsymbol{X}$ resides in the product Hilbert space $\mathcal{H} = L^2(\mathcal{T})^p$, of p-dimensional square-integrable functions [Jacques and Preda, 2014b, Wang et al., 2016]. The inner product and norm of $\boldsymbol{x} = (x_1, \ldots, x_p)'$ and $\boldsymbol{y} = (y_1, \ldots, y_p)'$ in $\mathcal{H}$ are given by

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle_{\mathcal{H}} = \sum_{j=1}^{p} \langle x_j, y_j \rangle = \sum_{j=1}^{p} \int_{\mathcal{T}} x_j(t) y_j(t) \mathrm{d}t \quad \text{and} \quad \|\boldsymbol{x}\|_{\mathcal{H}} = \langle \boldsymbol{x}, \boldsymbol{x} \rangle_{\mathcal{H}}^{1/2}, \tag{8}$$

respectively. A stochastic process is called an L^2 process if and only if it has finite second moments; $\mathbb{E}[\|\boldsymbol{X}(t)\|_{\mathcal{H}}^2] < \infty$ for all $t \in \mathcal{T}$. In particular, we consider L^2-continuous processes, i.e., L^2 processes for which $\lim_{h \to 0} \mathbb{E}[\|\boldsymbol{X}(t+h) - \boldsymbol{X}(t)\|_{\mathcal{H}}^2] = 0$ for every $t \in \mathcal{T}$, see, e.g., Ash and Gardner [2014] for more details. The covariance structure of $\boldsymbol{X}$ is represented by the continuous mean $\boldsymbol{\mu}(t)$ and matrix-variate covariance function/kernel $\boldsymbol{K}(s,t)$:

$$\boldsymbol{K}(s,t) = \begin{pmatrix} \kappa_{11}(s,t) & \cdots & \kappa_{1p}(s,t) \\ \vdots & \ddots & \vdots \\ \kappa_{p1}(s,t) & \cdots & \kappa_{pp}(s,t) \end{pmatrix}, \quad \kappa_{ij}(s,t) = \mathrm{Cov}(X_i(s), X_j(t)).$$

Since $\boldsymbol{X}$ is L^2-continuous, the corresponding covariance operator $\mathbf{C} : \mathcal{H} \to \mathcal{H}$, defined with $\mathbf{C}\,\boldsymbol{x}(s) = \int_{\mathcal{T}} \mathbf{K}(s,t)\boldsymbol{x}(t)\mathrm{d}t, \quad \boldsymbol{x} \in \mathcal{H}$, is a Hilbert-Schmidt operator, and the multivariate Mercer's theorem [Withers, 1974, Daw et al., 2022] implies that there exist countable sequences of continuous orthonormal eigenfunctions $\{\boldsymbol{\psi}_k\}_{k \geq 1}$ and non-negative decreasing eigenvalues $\{\pi_k\}_{k \geq 1}$ with $\sum_{k=1}^{\infty} \pi_k < \infty$ such that

$$\mathbf{C}\,\boldsymbol{\psi}_k = \pi_k\,\boldsymbol{\psi}_k \quad \text{and} \quad \mathbf{K}(s,t) = \sum_{k=1}^{\infty} \pi_k\,\boldsymbol{\psi}_k(s)\,\boldsymbol{\psi}_k'(t). \tag{9}$$

The multivariate Karhunen-Loève representation theorem then implies that there exists a unique sequence of uncorrelated random variables $\{\beta_k\}_{k \geq 1}$ such that

$$\boldsymbol{X}(t) = \boldsymbol{\mu}(t) + \sum_{k=1}^{\infty} \beta_k\,\boldsymbol{\psi}_k \quad \text{with} \quad \beta_k = \langle \boldsymbol{X} - \boldsymbol{\mu}, \boldsymbol{\psi}_k \rangle_{\mathcal{H}} = \int_{\mathcal{T}} \boldsymbol{\psi}_k'(t)(\boldsymbol{X}(t) - \boldsymbol{\mu}(t))\mathrm{d}t, \tag{10}$$

where $\beta_k \sim \mathcal{N}(0, \pi_k)$ if $\boldsymbol{X}$ is a multivariate Gaussian process [Daw et al., 2022].

In continuation, an L^2-continuous p-variate stochastic process $\boldsymbol{X}$ with mean function $\boldsymbol{\mu}$ and covariance function $\mathbf{K}$ is denoted as $\boldsymbol{X} \sim \mathcal{MSP}_p(\boldsymbol{\mu}, \mathbf{K})$.

Finite basis representation: In practice, we do not observe continuous functions but a discrete set of functional values. As discussed in Basna et al. [2022], a fundamental step in FDA is often to transform these discretely recorded data into a functional form, allowing each observed function to be evaluated at any point within its continuous domain $t \in \mathcal{T}$. Typically, the functional object is approximated by linear combinations of a finite number of basis functions, where this representation is

exact only for functions of finite rank.

Let $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$, $\boldsymbol{X}_i = [\boldsymbol{X}_{i,j}(t_k)]_{j=1,\ldots,p,\, k=1,\ldots,q} \in \mathbb{R}^{p\times q}$, be an i.i.d. sample of the multivariate random processes $\mathcal{MSP}(\boldsymbol{\mu}, \boldsymbol{\Sigma}^{\text{row}}, \kappa)$ with separable covariance $\mathbf{K} = \boldsymbol{\Sigma}^{\text{row}}\kappa$, observed at time points $t_1, \ldots, t_q \in \mathcal{T}$, for $q \in \mathbb{N}$. Given $m \in \mathbb{N}$, let $\boldsymbol{\phi}(t) = (\phi_1(t), \ldots \phi_m(t))'$, $t \in \mathcal{T}$ be a fixed basis that spans an m-dimensional subspace of $L^2(\mathcal{T})$. We transform each discretely observed $\boldsymbol{X}_i$, $i = 1, \ldots, n$, to a *functional form* by representing its components as a linear combination of basis functions in $\boldsymbol{\phi}$: $X_{i,j}^{(m)}(t) = \sum_{k=1}^{m} a_{j,k}^{(i)} \phi_k(t) = \boldsymbol{a}_j^{(i)\prime} \boldsymbol{\phi}(t)$, $j = 1, \ldots, p$, $i = 1, \ldots, n$, where superscript (m) emphasizes that $X_{i,j}^{(m)}$, $j = 1, \ldots, p$, $i = 1, \ldots, n$, is at most a *rank-m* process, meaning that its covariance operator has at most m non-zero eigenvalues.

Collecting all coefficients $\boldsymbol{a}_j^{(i)} = (a_{j,1}^{(i)}, \ldots, a_{j,m}^{(i)})'$, $j = 1, \ldots, p$, corresponding to the ith observation, in a matrix $\boldsymbol{A}_i = (\boldsymbol{a}_1^{(i)}, \ldots, \boldsymbol{a}_p^{(i)}) \in \mathbb{R}^{m\times p}$, we can represent each observation as

$$\boldsymbol{X}_i^{(m)}(t) = \boldsymbol{A}_i' \boldsymbol{\phi}(t), \quad i = 1, \ldots, n. \tag{11}$$

For simplicity of the notation, we drop the superscript (m) in the following and write $\boldsymbol{X}_i(t) = \boldsymbol{A}_i' \boldsymbol{\phi}(t)$.

The coefficients $\boldsymbol{a}_j^{(i)}$, $j = 1, \ldots, p$, $i = 1, \ldots, n$, are usually determined through least squares estimation, where the goal is to minimize the difference between the observed data and their approximation while ensuring smoothness; see Ramsay and Silverman [2005] for more details. Common choices of basis functions include splines, wavelets, and Fourier bases, among others; see, e.g., Ramsay and Silverman [2005] for an overview.

5.2 New developments

Notion of Mahalanobis distance: For univariate functional data, various approaches to outlier detection have been developed. A widely used method is the Mahalanobis distance, which measures the distance of a curve from a central tendency, accounting for variability [Galeano et al., 2015, Ghiglietti et al., 2017, Berrendero et al., 2020, Oguamalam et al., 2024a]. Robust alternatives, including depth-based methods, have also been proposed [Hubert et al., 2015].

In the multivariate functional setting, robust distance-based outlier detection remains underexplored. An exception is our work using a truncated Mahalanobis distance [Mayrhofer, 2024], but there is a need for alternatives. Non-parametric depth-based methods have been applied to multivariate functional data [Claeskens et al., 2014, López-Pintado et al., 2014, Nagy et al., 2016], however, often neglecting the time-correlation structure of the components of a multivariate random process. On the other hand, most of the mentioned approaches lack a formal extension of the Mahalanobis distance to account for multivariate functional relationships. For this purpose, the covariance operator $\mathbf{C}$ serves as the functional analog of the covariance matrix in multivariate statistics. Therefore, when defining the Mahalanobis distance for functional data in a manner analogous to the multivariate case, the inverse of the covariance operator plays a crucial role in the formulation. However, as a Hilbert-Schmidt operator, the covariance operator is, in general, not invertible, and a regularized covariance operator should be used instead. For univariate functional data, there have been several proposals on how to define a notion of Mahalanobis distance in an infinite-dimensional L^2 space: Galeano et al. [2015] introduced a method based on a spectral cutoff regularization, where the covariance operator is truncated to a finite number of components, making it invertible. Specifically, for a univariate stochastic process

$\mathcal{MSP}_1(\boldsymbol{\mu}, \kappa)$ its squared truncated functional Mahalanobis distance with truncation level $m \in \mathbb{N}$ is given by

$$\mathrm{fMD}^2(X, \mu; \kappa, m) = \mathrm{fMD}^2(X; m) = \sum_{i=1}^{m} \frac{1}{\lambda_i} \langle X - \mu, \xi_i \rangle^2, \tag{12}$$

where (λ_i, ξ_i), $i = 1, \ldots, p$, $\lambda_1 \geq \cdots \geq \lambda_m > 0$, denote the first m eigenpairs of the covariance operator $\mathcal{K}$ with kernel κ. This approach is computationally efficient and well-suited for smoothed functions represented by a finite basis. Ghiglietti et al. [2017] proposed an alternative method that introduces regularization through an additional parameter, offering greater flexibility and addressing convergence issues encountered in Galeano et al. [2015]. Berrendero et al. [2020] further extended the Mahalanobis distance definition by incorporating smoothing through reproducing kernel Hilbert spaces (RKHS), embedding regularization directly into the distance computation. Oguamalam et al. [2024b] further developed an efficient algorithm for robust estimation of regularized Mahalanobis distance [Berrendero et al., 2020]. Additionally, a regularized robust Mahalanobis distance for density functions has been proposed in Oguamalam et al. [2024a].

Using the decomposition (9), the notion of univariate functional MD (12) can be extended to a multivariate setting. For example, for $\boldsymbol{X} \sim \mathcal{MSP}(\boldsymbol{\mu}, \mathbf{K})$, $\boldsymbol{Y} \sim \mathcal{MSP}(\boldsymbol{\mu}, \mathbf{K})$ and $M \in \mathbb{N}$ such that $\pi_1 \geq \cdots \geq \pi_M > 0$, the squared truncated functional multivariate Mahalanobis semi-distance (fMMD) between $\boldsymbol{X}$ and $\boldsymbol{Y}$ (w.r.t. $\mathbf{K}$) is taken to be $\mathrm{fMMD}^2(\boldsymbol{X}, \boldsymbol{Y}; \mathbf{K}, M) = \sum_{k=1}^{M} \frac{1}{\pi_k} \langle \boldsymbol{X} - \boldsymbol{Y}, \boldsymbol{\psi}_k \rangle^2$, where $(\pi_k, \boldsymbol{\psi}_k)$ denotes the kth eigenpair of the covariance operator $\mathbf{C}$ with kernel $\mathbf{K}$, $k = 1, \ldots, m$, and $M \in \mathbb{N}$ determines the spectral cutoff. This extension accounts for both within-function and between-function correlations. This concept has been briefly discussed in Martino et al. [2019], primarily in the context of simulations and examples as a competitive method. Similarly, the squared truncated functional Mahalanobis semi-distance of $\boldsymbol{X} \sim \mathcal{MSP}_p(\boldsymbol{\mu}, \mathbf{K})$ (w.r.t. $\boldsymbol{\mu}$ and $\mathbf{K}$) is

$$\mathrm{fMMD}^2(\boldsymbol{X}; M) := \mathrm{fMMD}^2(\boldsymbol{X}, \boldsymbol{\mu}; \mathbf{K}, M) = \sum_{k=1}^{M} \frac{1}{\pi_k} \langle \boldsymbol{X} - \boldsymbol{\mu}, \boldsymbol{\psi}_k \rangle^2. \tag{13}$$

In his PhD thesis, Mayrhofer [2024] derived a connection between the randomness of $\boldsymbol{X}$ and the randomness of the chosen basis $\boldsymbol{A}$ by assuming separability of the covariance $\mathbf{K}$. This allowed for a connection between fMMD^2 of $\boldsymbol{X}$ and the squared matrix Mahalanobis distance of $\boldsymbol{A}$.

Robust covariance estimators for separable models: Unconstrained estimation of the covariance is often unattainable in practice. For example, in spatio-temporal models, the covariance structure must account for interactions between all spatial locations and time points, resulting in a covariance matrix that grows quadratically with the number of observed points. Estimating such a dense matrix requires large amounts of data and computational resources, and small sample sizes can lead to unstable or overfitted estimates. For these reasons, we will focus on the family of multivariate processes with separable covariance structure; for an overview, see, e.g., Chen et al. [2021]. For such processes, the kernel can be decomposed as $\mathcal{K}(s,t) = \mathbf{\Sigma}^{\mathrm{row}} \kappa(s,t)$, where $\mathbf{\Sigma}^{\mathrm{row}}$ models inter-component dependencies, and $\kappa(s,t)$ captures temporal covariance. We will particularly focus on a weighted sample covariance estimator for multivariate functional data. The weights will be determined based on the extended Mahalanobis distance, emphasizing observations that align closely with the central tendency while downweighting potential outliers. More formally, for the random sample $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n \sim \mathcal{MSP}_p(\boldsymbol{\mu}, \mathbf{K})$

the weighted sample covariance kernel will be defined as

$$\mathcal{K}_{\mathrm{w}}(s,t) = \frac{\sum_{i=1}^{n} w_i \left(\boldsymbol{X}_i(s) - \hat{\boldsymbol{\mu}}(s)\right)\left(\boldsymbol{X}_i(t) - \hat{\boldsymbol{\mu}}(t)\right)'}{\sum_{i=1}^{n} w_i},$$

where $w_i = f(\mathrm{fMMD}^2(\boldsymbol{X}_i, \hat{\boldsymbol{\mu}}; \hat{\mathbf{K}}))$ and f is a robust weight function based on some notion of multivariate Mahalanobis distance, and $\hat{\boldsymbol{\mu}}$ and $\hat{\mathbf{K}}$ are the robust estimates of the mean and the covariance, respectively.

Robust covariance estimators for non-separable models: The separability assumption is often made for convenience. While it simplifies covariance modeling for multivariate functional data, it does not fully capture real-world dependencies. Recent work [Aston et al., 2017, Masak and Panaretos, 2023] highlighted these limitations, and practical methods for non-separable structures were still lacking. New approaches, such as series expansions into separable components [Masak et al., 2023], have recently been proposed to address this. Tomas Masak (WU Wien) is one of our national collaborators; see Section 8. Thus, our goal is to develop robust covariance estimators for more general, non-separable models. A key challenge is maintaining robustness and efficiency when moving beyond simple separable structures. Estimation in fully unstructured models is often inherently inefficient, as the parameter space is too large for precise estimation. To ensure robustness, we need to carefully tailor the estimation to respect the underlying structural constraints of the model while still providing robustness. One approach we take to ensure robustness is via weighted covariance estimators, whose weights depend on robust Mahalanobis-type distances. Because distance estimation in these broader models is nontrivial, another central contribution of this task is the development of new robust distance estimators suited to these settings.

Clustering and classification: Clustering of functional data presents unique challenges that require careful methodological adaptations. As with other statistical frameworks, clustering techniques must be tailored to account for the infinite-dimensional nature of functional data. This is also similar for classification, where additional information is available on the class labels for the training data. The work of Jacques and Preda [2014a] provides an extensive guide to the different approaches to clustering within FDA, and Wang et al. [2024] reviews the literature on classification of functional data. Model-based clustering and classification is reviewed in Chamroukhi and Nguyen [2019] for univariate functional data, and a model-based clustering method for multivariate functional data is presented in Jacques and Preda [2014b]. Basically, clustering or classification procedures can be based on raw data that are typically observed on a finite, discrete grid. This leads to high-dimensional multivariate clustering/classification. During filtering, the curves are represented in a lower dimensional space. Here, it is common to express the functional data by either principal functions or another general basis of the underlying Hilbert space. Afterwards, clustering/classification is applied to the set of basis coefficients or the resulting principal scores.

In the clustering context, it is essential to select a distance metric that defines similarity between observations, as well as an appropriate notion of centrality for representing cluster centers. For example, Martino et al. [2019] proposed a clustering method for multivariate functional data based on the functional Mahalanobis distance of Ghiglietti et al. [2017]. Our goal is to use the robust regularized Mahalanobis distance as the basis for a distance metric in both clustering and classification. Functional data are typically sampled with noise and contain substantial redundancy, making some form of

regularization not only necessary but naturally interpretable as data filtering. Different regularization operators or strategies could therefore be advantageous for different tasks, yet this aspect is largely understudied. One key challenge of this task is to investigate and tailor such regularizations to achieve stable and meaningful clustering results. Several additional challenges arise. Robust clustering procedures generally require constraints, such as eigenvalue constraints in high-dimensional multivariate settings, to avoid degenerate solutions [García-Escudero and Mayo-Iscar, 2024]. It remains unclear which constraints are most suitable in the functional setting, especially for multivariate functions. For truncated representations, constraints based on eigenvalue size are possible, while more flexible regularization schemes could also prevent degeneracy and improve performance. A related challenge is constructing a robust pooled covariance estimator for classification. Direct pooling is unstable in high-dimensional or functional settings, making structural constraints necessary. Approaches based on structured covariance estimation [Cai et al., 2021] offer a promising foundation, and adapting them to the robust functional context forms another core component of this task.

Explainable clustering and classification for functional data: Explainable clustering and classification are typically treated in the ML community [Alvarez-Garcia et al., 2024], mainly in terms of visualizing the results. Mayrhofer and Filzmoser [2023] introduced the concept of Shapley values for explainability in connection with the use of a Mahalanobis distance as a distance metric. This turned out to be useful for a univariate square-integrable stochastic process $X \sim \mathcal{MSP}_1(\mu, \kappa)$ defined on a domain $\mathcal{T} \subseteq \mathbb{R}$, decomposed into disjoint subintervals $\mathcal{T} = \mathcal{T}_1 \cup \cdots \cup \mathcal{T}_d$, for which Mayrhofer [2024] developed Shapley contributions of the subdomains to the overall outlyingness. This framework was extended to multivariate processes $\boldsymbol{X} \sim \mathcal{MSP}_p(\boldsymbol{\mu}, \mathbf{K})$ and it has been shown how the overall outlyingness of the random process can be decomposed into coordinate-specific contributions, time-specific contributions, and time-coordinate-specific contributions for the disjoint subintervals $\mathcal{T}_a$ $(a = 1, \ldots, d)$ of $\mathcal{T}$ and coordinates $k = 1, \ldots, p$. This idea will serve as the basis for the development of explainability tools that reveal the reasons behind the class assignments of observations.

The following work package WP II addresses the research question formulated as hypothesis Hyp. II in Section 2. In particular, the objectives mentioned in Section 2, Obj. II.1–II.5, are addressed in the following tasks:

WP II **Robust regularized Mahalanobis distances for explainable classification and clustering of multivariate functional data**

Objectives: The purpose of this work package is to develop outlier-robust clustering and classification methodology for (univariate and) multivariate functional data. This includes the development of regularized Mahalanobis distances, of robust covariance estimators useful for classification and clustering, the derivation of theoretical properties, the implementation of efficient algorithms, and the development of tools for explainability.

Innovation: Almost no work is available on robust multivariate FDA, and we are at the forefront in developing Mahalanobis distance-based methodology for this purpose. Moreover, robust clustering and classification are new, and we are not aware of any tools for explainability in this context (not even for non-robust procedures).

(Task II.1) **Regularized Mahalanobis distance.** A regularized version of the robust Mahalanobis distance for multivariate functional data is defined. The initial ideas of Berrendero et al.

[2020], who proposed a regularized Mahalanobis distance for univariate functional data using ridge penalization, serve as a starting point. However, in the multivariate setting, ridge regularization alone is unlikely to be sufficient. Functional data are typically sampled with noise and redundancy, making regularization not only necessary but naturally interpretable as a form of data filtering. This perspective highlights that different regularization strategies may be preferable for different tasks, depending on the desired balance between smoothing, structural preservation, and sensitivity to relevant features. Consequently, a central focus of this work will be the exploration of more flexible, task-adapted regularization strategies, e.g., derivative-based penalties that control smoothness or curvature, and the investigation of their theoretical and practical implications in the multivariate functional setting. An accurate estimate of the covariance operator will be crucial.

(Task II.2) **Robust covariance estimators for separable and non-separable models.** Weights based on the regularized Mahalanobis distance from Task II.1 are used for a weighted sample covariance estimator. For non-separable multivariate functional data, a robust and computationally efficient framework is developed via both sequential robust estimation of the separable components, and the separable plus banded components, in the covariance expansion.

(Task II.3) **Distance-based classification and clustering methods.** A truncated version of the functional Mahalanobis (semi-)distance [Mayrhofer, 2024] or, preferably, the regularized distance developed in Task II.1 is used for robust clustering and classification. The performance of classification and clustering procedures based on different variants of distances and covariances is investigated and compared, both theoretically and empirically. Different types of constraints for robust clustering are evaluated, and robust pooling versions are developed and compared.

(Task II.4) **Tools to obtain explainability for classification and clustering.** We will focus on explainability w.r.t. the individual functional observations in terms of identifying the attributes that contribute most to the group membership association. Especially for Mahalanobis distance-based procedures, our Shapley value framework for functional outlier detection seems very appropriate for explainability in this context. However, a solid theoretical basis for the theoretical properties of Shapley values in a k-group setting has to be derived. Visualization tools need to be developed that efficiently display the coordinate/time-specific contributions to group membership. Our experience with developing explainability tools for clustering/classification in the matrix- and tensor-valued setting will be very helpful here.

(Task II.5) **Theoretical properties of the estimators.** Robustness properties such as the breakdown point [Maronna et al., 2019] are difficult to derive in this context. In our work Puchhammer et al. [2025], we derived the breakdown point for a Gaussian mixture model setting for vector-valued data, which can serve as a good starting point. The distributional properties of the robust covariance estimators and the regularized Mahalanobis distance will be studied under various stochastic process assumptions, including separability and ellipticity.

6 Timeline and dissemination plan

This section provides an overview about the work packages and tasks. The two work packages address the hypotheses Hyp. I and Hyp. II formulated in Section 2, and the tasks are directly connected to the objectives mentioned there. An overview on the timeline is presented in Figure 4. The lead of WP I is jointly taken by PF and MM, and the responsible persons for WP II are PF and UR. We have a very good experience with our joint collaborations. As all team members are working in the same unit, with offices next to each other, there will be an exchange even on a daily basis, and naturally a very close collaboration will result. Therefore, each team member will contribute to each task.

PF will primarily be responsible for project management and detailed work planning, as well as for coordinating meetings and exchanges with national and international collaboration partners (see Section 8). Such meetings will be held on a regular basis. In addition, we will have "literature" meetings to jointly present recent papers on the project topics. We also plan to invite experts in the field to TU Wien for seminar presentations and possible collaborations.

The project results will be presented at national and international conferences and published as open-access publications in highly ranked journals. The developed software will be published as open source software in the form of R packages. Related R packages that we have already published are Oguamalam et al. [2023] and Mayrhofer et al. [2025b]. A project web page will provide an immediate overview of our achievements.

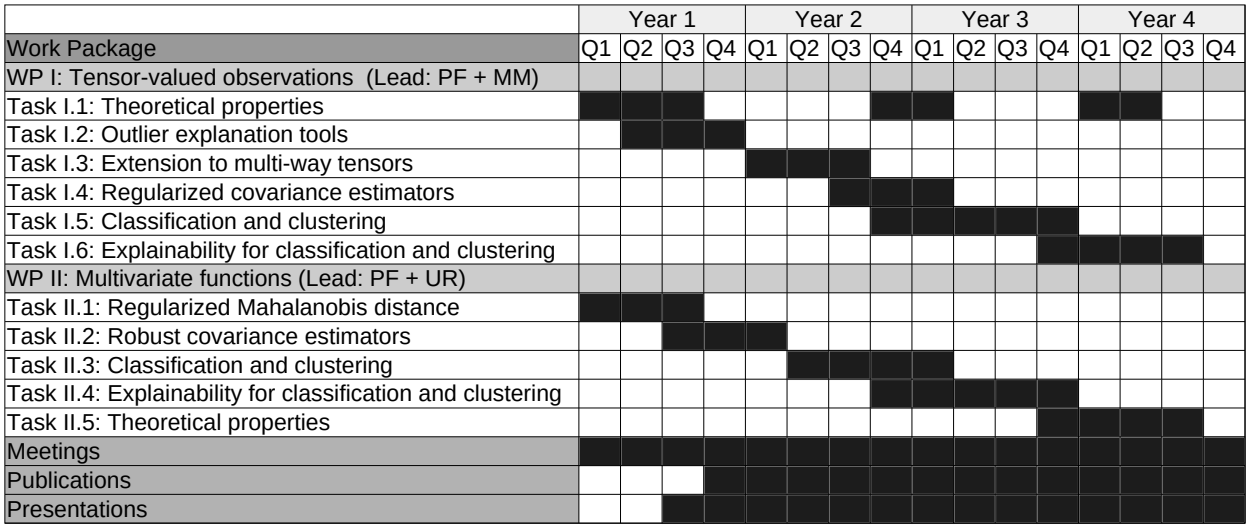


Figure 4: Work packages, responsibilities, tasks, and timeline.

7 Expected implications for other areas

Nowadays, data collection is done in practically all areas, and their structure is becoming increasingly complex: measurements may be recorded at high frequency or high resolution, include additional contextual information, vary across time, or take the form of images, videos, or other high-order objects. The methods and technology developed in this project address some of the challenges, particularly those where the information per subject is not just a vector but an object of higher order. Processing this type of information directly, rather than vectorizing it, enables a more accurate extraction of

the inherent information. Since the aim is to develop robust methods, data artifacts, outliers, or inconsistencies in the measurements will be accounted for during the analysis. These developments have broad practical value, and can be particularly useful for medical image analysis, where the outlier information can help identify unusual patterns relevant for diagnosis, but also in other areas where image or video information is recorded. E.g., in engineering, they can support fault detection using sensor data. Similar benefits apply to fields that rely on spectral or longitudinal data, such as e.g. chemometrics or medicine. These data are prone to errors due to factors like measurement processes or sample preparation variability. We will demonstrate the effectiveness of our methods using real-world datasets and collaborate closely with domain experts to ensure the practical relevance of our findings.

8 National and international cooperation arrangements

We will emphasize international collaborations to ensure active knowledge transfer of our current research. Specifically, we plan to work with experts in the field:

Tomas Masak (WU Wien, Austria): He is an expert in Functional Data Analysis, for both curve and surface-valued data. In case of a two-dimensional domain he extended the restriction of separability for the covariance operator. He developed theory, but also computationally efficient algorithms making the proposals accessible for practical applications. He publishes in top journals such as Journal of Computational and Graphical Statistics, Biometrika, or the Journal of the American Statistical Association. As an expert in FDA, he will contribute primarily to the FDA in Tasks II.1 and II.2. in WP II. We will particularly collaborate on developing more general models when the separability assumption does not hold. Within these non-separable models, we will study the explainable outlier detection from both theoretical and computational perspectives.

Mia Hubert (KU Leuven, Belgium): She is a key researcher in Robust Statistics, with various contributions in the field of multivariate statistics, high-dimensional data analysis, and also Functional Data Analysis. She has numerous publications in top journals. As an expert in robust methodology, she will contribute to the conceptualization and implementation of the novel methodology in Task I.5 and Task II.3.

Joni Virta (University of Turku, Finland): He is an expert with matrix- and tensor-valued data, and he works in the general domain of dimension reduction techniques. He has an impressive publication record in high-level journals and also has a long-term collaboration with Una Radojičić. He will primarily contribute to the tensor data analysis in WP I. Building on the previous work of U. Radojičić and J. Virta on latent dimension estimation for tensorial data, we plan to extend this collaboration by developing robust methods for estimating the underlying signal dimension. We will also explore feature extraction techniques using robust principal and independent component analysis, as well as projection pursuit.

9 Human resources

The research will be carried out by team members of the Computational Statistics (CSTAT) group led by Prof. P. Filzmoser. CSTAT is part of the Institute of Statistics and Mathematical Methods in Economics (ISW) at TU Wien. The available infrastructure of CSTAT is outlined in Annex 2.

Peter Filzmoser is a Full Professor in Statistics. His work in Robust Statistics is very well recognized

in the statistics community already since many years. Recently, he collaborated with Una Radojičić and Marcus Mayrhofer on developing robust methods for matrix-variate and functional data. Also this work already received a lot of attention at international conferences. Within this project he will actively contribute to method development for all mentioned tasks. He will take the lead in this project in order to fulfill the project goals according to the timeline mentioned in Section 6. He will be the main supervisor of the PhD student employed in this project. Publications relevant to the project are Puchhammer et al. [2025, 2026] as well as those listed for UR and MM below.

Una Radojičić is a Post-Doc at the CSTAT group. She finished her doctoral study in the CSTAT group in 2021, which was focused on non-Gaussian feature extraction for complex data. Since 2022 she is involved in a project devoted to functional data analysis in the context of compositional data. Una has a strong Mathematics background, and a lot of expertise in multivariate statistics, functional data analysis, compositional data analysis, robust statistics, and statistical computing. Una will contribute to all project tasks for the complete duration of the project, and she will co-supervise the PhD student. She will use part of the project funding, especially in the first half of the project, with a 50% PostDoc position (plus a small overlap period). Publications relevant to the project are Oguamalam et al. [2024a,b] and Mayrhofer et al. [2025a].

Marcus Mayrhofer is a Post-Doc at the CSTAT group. He finished his doctoral study in the CSTAT group in 2024, where he introduced the concept of Shapley values for outlier explanation, as well as robust statistical tools for matrix-variate observations. With his strong background in Mathematics and Statistics, he will also contribute to all project tasks and co-supervise the PhD student. He will use part of the project funding, especially in the second half of the project, with a 50% PostDoc position (plus a small overlap period), but contribute continuously during the 48 months. Publications relevant to the project are Mayrhofer and Filzmoser [2023], Mayrhofer et al. [2025a].

PhD student N.N. During the project, it is planned to hire a PhD student for the full project duration. We opted for a duration of 48 months rather than 36 months, as this will give us time to develop high-quality contributions, and thus papers in highly ranked journals. The PhD student should have a master's degree in statistics or mathematics with a focus on statistics.

10 Ethical, safety-related, or regulatory aspects

In general, no ethical permits are required for this project. The research questions and results do not contain ethical aspects. There are also no obvious dual-use (military or intelligence) aspects. If ethical safety-related aspects become relevant (e.g., when collaborating with applied scientists), we will consult the department for Responsible Research Practices at TU Wien, and obey the ethical standards and the data protection regulation at TU Wien.

11 Sex-specific, gender- & diversity-related aspects

We did not identify any relevant sex-specific, gender- or diversity-related aspects in the research of this project. If such aspects become relevant (e.g., when collaborating with applied scientists or considering datasets with sex-specific, gender- or diversity-related aspects), we will consult the department Gender Equality Office at TU Wien and obey the standards of TU Wien.

A Annex 1: References

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B Annex 2: Lead institution and financial aspects

B.1 Existing personnel and infrastructure

The Institute of Statistics and Mathematical Methods in Economics (ISW) at TU Wien has eight research units. This project will be placed at the Computational Statistics[1] (CSTAT) research group, led by Prof. Peter Filzmoser, which is very active in developing statistical methods and analyzing data. The group has two senior researchers and several postdocs and PhD students. The group's main focus lies in Robust Statistics and Compositional Data Analysis, with Peter Filzmoser as a core researcher in this area. The group has many international contacts with top researchers in various fields. The CSTAT group also has eminent expertise with complex data structures, such as high-dimensional, functional, and higher-order data. Another focus of the CSTAT research group is to actively collaborate with more applied sciences at an international as well as local (TU Wien and Austrian research groups) level (e.g., computer science, visualization, chemical engineering, civil engineering, geodesy, etc.) as there is a great demand for support on statistical data analysis tasks, connected with the development of new methods. The CSTAT group also provides the computationally needed hardware, workstations, and other necessary equipment. The following few projects from the last years further outline the expertise of CSTAT:

- SEMACRET - Sustainable exploration for orthomagmatic (critical) raw materials in the EU: Charting the road to the green energy transition
- DGBIAS – Detecting gender bias in children's books (FWF)
- Generalized relative data and robustness in Bayes spaces (FWF)
- AI4CSM - Automotive intelligence for connected shared mobility (EU), with many international scientific and company partners

In total, CSTAT provides a motivating and stimulating research environment.

B.2 Cost justifications of requested funding

Table 1 summarizes the requested funding. The subsequent chapters explain the details for each item.

Table 1: Total requested funding

Item	Amount
Personnel	419.314,01 €
Travel	9.000,00 €
Equipment, Material, Other	0,00 €
Subtotal	428.314,01 €
General project costs (5%)	21.415,70 €
Total	**449.729,71 €**

[1] https://www.tuwien.at/en/mg/cstat

B.2.1 Personnel costs (Total: 419.314,01 €)

The personnel costs fund the following positions:

N.N. This position is for a PhD student (30 hours) for the duration of 48 months. The PhD student needs to have a strong background in Mathematics, with a focus on Statistics. We have access to potential candidates at TU Wien and even regular requests from our students to start a PhD. However, we will also announce this position at other national and international institutions.
Job description: The PhD student will be majorly involved in all project tasks, thus to the work packages I and II. The tasks consist of literature review, developing new methods, implementation of methods, deriving theoretical properties, and working on publications.
We aim to achieve high-level scientific publications, including significant theoretical contributions. Given the time-intensive nature of this process, we have opted for a 48-month project duration. The PhD student will be supervised by Peter Filzmoser and co-supervised by Una Radojičić and Marcus Mayrhofer.

Una Radojičić will be employed as a PostDoc for 50% in the first 24 months, and then with 10% for the third year. We will ensure that she receives funding from other sources throughout the project period so that she can contribute fully to the project for the entire duration.
Job description: She will contribute with theoretical and algorithmic developments to all project tasks, as well as to paper writing.

Marcus Mayrhofer will be employed as a PostDoc for 50% in years 3 and 4, and with 10% in the second year. We will ensure that he receives funding from other sources throughout the project period so that he can contribute fully to the project for the entire duration.
Job description: He will contribute with theoretical and algorithmic developments to all project tasks, as well as to paper writing.

B.2.2 Travel costs (Total: 9.000,00 €)

The travel budget is intended for visits to our international collaborators Joni Virta (Finland) and Mia Hubert (Belgium) for an intensive collaboration to develop the methods described in work packages I and II, and also for preparing joint publications. We will use virtual meetings for an exchange as much as possible, but we strongly believe that visits to the partners are very useful, if not crucial, for the project progress.

Table 2 provides details about the requested travel funding: Una Radojičić will spend one week at the beginning of the project with Joni Virta at the University of Turku to initialize research on tensor-valued observations and to outline possible directions. Similarly, Marcus Mayrhofer will visit Mia Hubert at KU Leuven at the beginning of the project in order to explore possible directions for method development for function data. The PhD student will spend 3 visits in Turku and at KU Leuven to align our research directions with the ideas of the partners and to get new input from them. Mayrhofer will also visit Virta, and Radojičić will visit Hubert to coordinate the research.

In Table 2, 1 week refers to a stay of 5 days (4 nights), resulting in 650 € of accommodation costs. For travel expenses to Turku we calculated 300 €, and to Leuven we calculated 200 €.

Table 2: Total requested travel funding

Person	Destination	Time	Duration	Cost
Radojičić	Univ. Turku	Y1, Q1	1 week	950 €
PhD student	Univ. Turku	Y1, Q3	1 week	950 €
PhD student	Univ. Turku	Y2, Q3	1 week	950 €
Mayrhofer	Univ. Turku	Y3, Q2	1 week	950 €
PhD student	Univ. Turku	Y3, Q3	1 week	950 €
Mayrhofer	KU Leuven	Y1, Q1	1 week	850 €
PhD student	KU Leuven	Y2, Q1	1 week	850 €
PhD student	KU Leuven	Y3, Q1	1 week	850 €
Radojičić	KU Leuven	Y3, Q2	1 week	850 €
PhD student	KU Leuven	Y4, Q1	1 week	850 €
Total				**9.000 €**

B.2.3 Equipment, material and other costs

Not applicable in this project.

C Annex 3: Scientific CVs and achievements

C.1 CV Peter Filzmoser

Degrees:	Univ.-Prof. Dipl.-Ing. Dr.techn.
Current position:	Full Professor "Statistics"
Address:	TU Wien (Vienna University of Technology)
	Institute of Statistics and Mathematical Methods in Economics
	Research Group *Computational Statistics*
	Wiedner Hauptstraße 8-10, A–1040 Vienna, Austria
Office phone:	+43 1 58801 10560
E-mail:	Peter.Filzmoser@tuwien.ac.at
WWW:	https://www.tuwien.at/mg/cstat/team/filzmoser
ORCID:	0000-0002-8014-4682

Academic Milestones

2001: Habilitation in Statistics, TU Wien.
1996: Doctor of Technical Sciences, TU Wien.
1993: M.Sc. in Mathematics, TU Wien.

Academic Positions

1993–2001: Assistant Professor, Department of Statistics and Probability Theory, TU Wien.

March–July 2005: Visiting Professor, Department of Statistics and Decision Support Systems, University of Vienna.

January and June 2007: Visiting Professor, GREMAQ (Groupe de Recherche en Economie Mathématique), Université de Toulouse I, France.

March–May 2008: Visiting Professor, Department of Applied Mathematics, Belarusian State University, Minsk, Belarus.

2012–2015: Consultant for Palacký University Olomouc, Czech Republic (1/2 position)

2001–2017: Associate Professor, Department of Statistics and Probability Theory, TU Wien.

2017–present: Full Professor, Institute of Statistics and Mathematical Methods in Economics, TU Wien.

Areas of Research Peter Filzmoser has conducted research in the area of Robust Statistics, including outlier detection, Multivariate Statistics, and Compositional Data Analysis. He contributed to the development of new methods and algorithms for complex data, including high-dimensional data, functional data, and data depending on space/time. He is interested in theoretical properties of the methods, as well as in applications in various different fields.

Publication Overview

A complete list of publications is available at:
https://www.tuwien.at/mg/cstat/publikationen/filzmoser-peter

More than 300 peer reviewed publications, mainly in scientific journals; co-author of 3 books (Springer, Wiley, CRC Press)

Top Ten Publications:

1. M. Mayrhofer, U. Radojičić, and P. Filzmoser. (2025) Robust covariance estimation and explainable outlier detection for matrix-valued data. *Technometrics*, 67(3):516–530.
   DOI: 10.1080/00401706.2025.2475781
2. P. Pfeiffer, A. Alfons, and P. Filzmoser. (2025) Efficient computation of sparse and robust maximum association estimators. *Computational Statistics & Data Analysis*, 207, 108133.
   DOI: 10.1016/j.csda.2025.108133
3. L. Neubauer and P. Filzmoser. (2024) Improving forecasts for heterogeneous time series by "averaging", with application to food demand forecast. *International Journal of Forecasting*, 40(4), 1622-1645.
   DOI: 10.1016/j.ijforecast.2024.02.002
4. J. Oguamalam, U. Radojicic, and P. Filzmoser. (2024) Minimum regularized covariance trace estimator and outlier detection for functional data. *Technometrics*, 66(4), 588-599.
   DOI: 10.1080/00401706.2024.2336542
5. F.S. Kurnaz and P. Filzmoser. (2023) Robust and sparse multinomial regression in high dimensions. *Data Mining and Knowledge Discovery*, 37, 1609-1629.
   DOI: 10.1007/s10618-023-00936-6
6. M. Mayrhofer and P. Filzmoser. (To appear) Multivariate outlier explanations using Shapley values and Mahalanobis distances. *Econometrics and Statistics*.
   DOI: 10.1016/j.ecosta.2023.04.003
7. P. Puchhammer and P. Filzmoser. (2023) Spatially smoothed robust covariance estimation for local outlier detection. *Journal of Computational and Graphical Statistics*, 33(3), 928–940.
   DOI: 10.1080/10618600.2023.2277875
8. A. Alfons, C. Croux, and P. Filzmoser.(2017) Robust maximum association estimators. *Journal of the American Statistical Association*, 112 (517), 436-445.
   DOI: 10.1080/01621459.2016.1148609
9. C. Croux, P. Filzmoser, and H. Fritz. (2013) Robust sparse principal component analysis. *Technometrics*, 55(2), 202-214.
   DOI: 10.1080/00401706.2012.727746
10. P. Filzmoser, R. Maronna, and M. Werner. (2008) Outlier Identification in high dimensions. *Computational Statistics and Data Analysis*, 52, 1694-1711.
    DOI: 10.1016/j.csda.2007.05.018

Additional Research Achievements (Ten Most Relevant Ones)

1. Supervised 25 PhD students
2. Main organizer of 10 international scientific conferences
3. Scientific (co-)chair of international conferences (ICORS, DSSV, Compstat, CodaWork)
4. 230 scientific presentations at conferences and research institutions; 12 were Keynote, 175 were Invited

5. Leadership roles in several professional societies (International Association for Statistical Computing – IASC, Austrian Statistical Society)
6. Elected Member of the International Statistical Institute (ISI)
7. Felix Chayes Prize awarded from the International Association for Mathematical Geosciences (IAMG) for Excellence in Research in Mathematical Petrology (2019)
8. Various research grants (4 x FWF, 4 x EU, 7 x FFG, 1 x OeAD)
9. Member of several editorial boards (Journal Computational Statistics; JDSSV – Journal of Data Science, Statistics, and Visualisation; AStA – Advances in Statistical Analysis; Austrian Journal of Statistics; Chemometrics and Intelligent Laboratory Systems)
10. Contributing with several software packages to the statistical software environment `R`

C.2 CV Una Radojičić

Degrees:	Mag. Dr.
Current position:	PostDoc Project Assistant
Address:	TU Wien (Vienna University of Technology)
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ORCID:	0000-0003-0329-0595

Academic Milestones

2023: Habilitation for docent status in Statistics, University of Slavonski Brod, Croatia
2021: Doctor of Natural Sciences, TU Wien.
2017: Master of Mathematics, J.J. Strossmayer University of Osijek, Croatia.

Academic Positions

2025 – present: PostDoc University Assistant, Institute of Statistics and Mathematical Methods in Economics, TU Wien.

2021 – 2025: PostDoc Project Assistant, Institute of Statistics and Mathematical Methods in Economics, TU Wien.

2019 – 2021: PreDoc Project Assistant, Institute of Statistics and Mathematical Methods in Economics, TU Wien.

2018: Visiting researcher, Yale University.

2017 – 2019: Teaching assistant at Josip Juraj Strossmayer University of Osijek, Croatia.

Areas of Research Una Radojičić has conducted research in the fields of Multivariate, Computational, Functional, and Robust Statistics. Her work includes the development of robust covariance estimation methods for matrix-valued data and frameworks for functional outlier detection. She has contributed to the field by advancing methodologies for dimension reduction, blind source separation, and two-dimensional principal component analysis, establishing theoretical foundations while addressing practical applications in statistical modelling.

Publication Overview

A complete list of publications is available at:
https://www.tuwien.at/mg/cstat/publikationen/radojicic-una

Top Ten Publications:

1. U. Radojičić, K. Nordhausen, and J. Virta. (To appear) Kurtosis-based projection pursuit for matrix-valued data. *The Annals of Statistics*.

https://www.e-publications.org/ims/submission/AOS/user/submissionFile/66579?confirm=713aa259

2. U. Radojičić, J. Virta. (2025) Dimension estimation in PCA model using high-dimensional data augmentation. *Biometrika*.
   DOI: 10.1093/biomet/asaf052
3. U. Radojičić, N.Lietzén, K.Nordhausen, & J.Virta (2025) Order Determination for Tensor-valued Observations Using Data Augmentation. *Journal of Computational and Graphical Statistics*, 1–20.
   DOI: 10.1080/10618600.2025.2500977
4. M. Mayrhofer, U. Radojičić, and P. Filzmoser. (2025) Robust covariance estimation and explainable outlier detection for matrix-valued data. *Technometrics*, 67(3), 516–530.
   DOI: 10.1080/00401706.2025.2475781
5. Oguamalam, J. Radojičić, U. and Filzmoser, P. (2024). Minimum regularized covariance trace estimator and outlier detection for functional data. *Technometrics*, 66(4), 588-599.
   DOI: 10.1080/00401706.2024.2336542
6. Radojičić, U., Nordhausen, K., & Virta, J. (2025) Unsupervised linear discrimination using skewness. *Journal of Multivariate Analysis*.
   DOI: 10.1016/j.jmva.2025.105524
7. Archimbaud, A., Drmač, Z., Nordhausen, K., Radojičić, U., & Ruiz-Gazen, A. (2023) Numerical Considerations and a new implementation for invariant coordinate selection. *SIAM Journal on Mathematics of Data Science*, 5(1), 97-121.
   DOI: 10.1137/22M1498759
8. Radojičić, U., Nordhausen, K., & Virta, J. (2021) Large-sample properties of unsupervised estimation of the linear discriminant using projection pursuit. *Electronic Journal of Statistics*, 15(2), 6677-6739.
   DOI: 10.1214/21-EJS1956
9. Grygar, T. M., Radojičić, U., Pavlů, I., Greven, S., Nešlehová, J. G., Tůmová, Š., & Hron, K. (2024) Exploratory functional data analysis of multivariate densities for the identification of agricultural soil contamination by risk elements. *Journal of Geochemical Exploration*, 259, 107416.
   DOI: 10.1016/j.gexplo.2024.107416
10. Radojičić, U., Nordhausen, K., & Oja, H. (2020) Notion of information and independent component analysis. *Applications of Mathematics*, 65(3), 311-330.
    DOI: 10.21136/AM.2020.0326-19

Additional Research Achievements

1. Invited speaker at several international conferences, including CMStatistics, ICORS, DSSV, SMPS, ICSDS, and others.
2. Reviewer for Journal of Computational and Graphical Statistics and International Statistical Review.
3. Receiver of the Christiana Hörbiger prize of the TU Wien in 2020.

C.3 CV Marcus Mayrhofer

Degrees:	Dipl.-Ing. Dr.techn.
Current position:	PostDoc University Assistant
Address:	TU Wien (Vienna University of Technology)
	Institute of Statistics and Mathematical Methods in Economics
	Research Group *Computational Statistics*
	Wiedner Hauptstraße 8-10, A–1040 Vienna, Austria
Office phone:	+43 1 58801 105687
E-mail:	marcus.mayrhofer@tuwien.ac.at
ORCID:	0000-0002-3430-8308

Academic Milestones

2024: Doctor of Technical Sciences, TU Wien.

2021: M.Sc. in Mathematics, TU Wien.

Academic Positions

2024 – present: PostDoc University Assistant, Institute of Statistics and Mathematical Methods in Economics, TU Wien.

2021 – 2024: PreDoc Project Assistant, Institute of Statistics and Mathematical Methods in Economics, TU Wien.

Areas of Research Marcus Mayrhofer has conducted research in the fields of Multivariate, Computational, Functional, and Robust Statistics. His research includes the development of frameworks for explainable outlier detection and highly robust matrix-variate covariance estimation. He has contributed to the field by creating publicly accessible algorithms and establishing theoretical foundations for these methods, bridging theory and practical application.

Publication Overview

1. M. Mayrhofer, U. Radojičić, and P. Filzmoser. (2025) Robust covariance estimation and explainable outlier detection for matrix-valued data. *Technometrics*, 67(3), 516–530. DOI: 10.1080/00401706.2025.2475781
2. M. Mayrhofer and P. Filzmoser. (2023). Multivariate outlier explanations using Shapley values and Mahalanobis distances. *Econometrics and Statistics*. DOI: 10.1016/j.ecosta.2023.04.003
3. Mayrhofer, M., Lewitschnig, H., & Filzmoser, P. (2025). A New View to Mission Profiles. *2025 Annual Reliability and Maintainability Symposium (RAMS)* (pp. 1-6). IEEE. DOI: 10.1109/RAMS48127.2025.10935003

A complete list of publications is available at:
https://www.tuwien.at/mg/cstat/publikationen/marcus-mayrhofer

Published Software

1. Mayrhofer, M., Radojicic, U., and Filzmoser, P. (2024). *robustmatrix: Robust Matrix-Variate Parameter Estimation.* R package version 0.1.4
2. Mayrhofer, M. and Filzmoser, P. (2022). *ShapleyOutlier: Multivariate Outlier Explanations using Shapley Values and Mahalanobis Distances.* R package version 0.1.2

Additional Research Achievements

1. Award for best dissertation in theoretical statistics in 2025 of the Austrian Statistical Society
2. Industry research project *Cellwise robust outlier detection for production data*, with the support of voestalpine Stahl GmbH, Linz, Austria (2025-26)
3. Speaker at more than ten international conferences, with invited contributions to COMPSTAT, ICORS, and CLADAG
4. Reviewer for the Journal *International Statistical Review*
5. Coordinator of the doctoral college *Innovative Combinations and Applications of AI and ML* (iCAIML)
6. (Co-)supervision of 3 bachelor's and 4 master's theses
7. Performance scholarship of TU Wien, Austria (academic years 2019-2020 and 2020-2021)

D Annex 4: Collaboration letters

D.1 Tomas Masak, WU Wien, Austria

WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

RNDr. Tomas Masak, D.Sc.
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Phone: +43-1-313 36 4308
E-mail: tmasak@wu.ac.at

To whom it may concern:
A letter of collaboration
Vienna, November 6th, 2025

With this letter I express my support and strong interest in being a scientific collaborator on the proposed research project titled "Robustness and explainability for complex data objects", submitted to the Austrian Science Fund FWF under Peter Filzmoser as the principal investigator.

My research in functional data analysis, namely on multivariate and multidimensional functional data, fits perfectly into the scope of the proposed project. Thus I am looking forward to the outcome of the joint collaboration.

Sincerely yours,



Tomas Masak

D.2 Mia Hubert, KU Leuven, Belgium

KU LEUVEN

FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS
SECTION OF STATISTICS AND DATA SCIENCE
CELESTIJNENLAAN 200B, BOX 2400
3001 LEUVEN, BELGIUM

To the Austrian Science Fund FWF

OUR REFERENCE
YOUR REFERENCE
LEUVEN November 5, 2025

Letter of collaboration

With this letter I express my support and strong interest in being a scientific collaborator on the proposed research project titled "Robustness and explainability for complex data objects", submitted to the Austrian Science Fund FWF under Peter Filzmoser as the principal investigator.

I am working in the field of Robust Statistics, with contributions also to Functional Data Analysis, and therefore I am looking forward to the joint research on new and innovative developments in this area.

With friendly greetings



Mia Hubert

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PHONE + 32 16 32 20 23
e-mail: mia.hubert@kuleuven.be
wis.kuleuven.be/statdatascience/robust

D.3 Joni Virta, University of Turku, Finland

UNIVERSITY OF TURKU

Joni Virta, PhD, Academy Research Fellow, Assistant Professor
Department of Mathematics and Statistics
University of Turku
joni.virta@utu.fi

Letter of Collaboration

With this letter I express my support and strong interest in being a scientific collaborator on the proposed research project titled "*Robustness and explainability for complex data objects*", submitted to the Austrian Science Fund FWF under Peter Filzmoser as the principal investigator.

My research on the analysis of matrix and tensor data is in perfect alignment with the topics mentioned in the proposal, and thus I am looking forward to the collaboration. I know the CSTAT team members quite well already due to several visits to them and also through collaborations with Una Radojičić.

Sincerely,



Joni Virta
Turku, Finland
5th of November 2025